0. Submission intent

This is a **V0.3.4 pipeline-test** submission for the PENGWIN 2026 Task 1 preliminary phase. The objective is to validate the end-to-end Grand Challenge contract (Docker container build, model tarball, `/input` MHA read, `/output` MHA write under `--network none`) with the current best in-house two-stage pipeline. It is **not** a competitive scientific submission and should not be interpreted as our final leaderboard entry. The femur anatomy (label range 151–200) is **intentionally not modeled** in V0.3.4 — femur fragments are emitted as background (0). Femur modeling is the V0.5 milestone (in progress).

1. Method overview

V0.3.4 is a **two-stage anatomy-conditioned pipeline** wrapped in a **4-layer robust pipeline** for Grand Challenge out-of-distribution defense.

- **Stage 1** is a whole-CT anatomy classifier (nnU-Net v2 ResEnc-L on `Dataset532_PelvicAnatomyV2`) that produces a 4-class softmax over background / Sacrum / LeftHipbone / RightHipbone.
- **Stage 2** is a per-anatomy V291 BICM nnU-Net (boundary-weighted trainer) on `Dataset537_PelvicBICMFragmentV5` that takes a **3-channel ROI** — bone-LUT CT, Stage 1 anatomy probability, and a signed distance map clipped to ±40 mm — and produces a 5-class V5 BICM softmax (background / exterior_context / interior_shell / core / contact_surface).
- **Decoder** is the V288-style core-seed watershed: it lifts the V5 softmax to per-instance fragment IDs using the core class as seeds and a 12× ridge cost on the contact class. Followed by a ~1 cm^3 connected-component prune and a ≤27-voxel small-fragment merge.
- **V0.3.4 4-layer robust pipeline**: - L1) bone-skeleton anatomy decomposition (HU > 200 + 3D CC) — distribution-independent fallback - L2) Ds532 argmax masks — mutually-exclusive anatomy assignment - L3) post-pad bbox sanity (≤ 50% of volume) — rejects out-of-distribution masks - L4) 480-second time budget — guarantees the 10-minute GC limit

2. Stage 1 — Ds532 anatomy classifier

- **Architecture**: nnU-Net v2 ResEnc-L (`nnUNetResEncUNetLPlans`, `3d_fullres`), 1-channel input, 4-class softmax.
- **Training data**: `Dataset532_PelvicAnatomyV2` (170 pelvic-only cases). Validation Dice 0.9684.
- **Input normalization**: LPS canonicalize + bone HU window [–1000, 2000] clip + bone-LUT normalize.
- **Output**: full-CT 4-class softmax over background / Sacrum / LeftHipbone / RightHipbone.

3. Per-anatomy ROI extraction

For each of {Sacrum, LeftHipbone, RightHipbone}:

1. Compute the bounding box of the Stage 1 `prob ≥ 0.5` mask, padded by 24 voxels on each axis. 2. Apply Layer 3 (bbox sanity): if `post_pad_bbox / volume > 0.5`, fall back to the bone-skeleton mask from Layer 1. 3. Build a 3-channel ROI: - bone-LUT normalized CT crop - Stage 1 anatomy probability crop - signed distance to the `prob ≥ 0.5` surface, clipped to ±40 mm

4. Stage 2 — V291 BICM nnU-Net

- **Architecture**: nnU-Net v2 ResEnc-L, 3-channel input, 5-class softmax (V5 BICM: bg/exterior/shell/core/contact).
- **Trainer**: `PengwinTrainerBoundaryFragmentSidecarCoreRecallDenseCandidateCore025StrongPeakNoContactABBCV291` — V288 ABBC base with 5× weight on the boundary class CE+Dice terms.
- **Training data**: `Dataset537_PelvicBICMFragmentV5` (12 of 174 cases, single fold, fold0). Phase 6 fulltrain (all 170 pelvic cases) reached EMA 0.64 at epoch 71 before being stopped for the V0.x restructure.
- **Sidecars**: V5 instance + BFv3 target (V291 dataloader contract).

5. V288 core-seed watershed decoder

- Use V5 class-3 (core) as seeds for `scipy.ndimage.label`.
- Use V5 class-4 (contact_surface) as a 12× ridge cost.
- Run `skimage.segmentation.watershed` with `mask = support` (`support = {2, 3, 4}`).
- Per-fragment connected-component prune at ~1 cm^3.

• Small-fragment merge at ≤ 27 voxels (V0.3.4).

6. Label assembly

Per-anatomy fragment IDs are offset into the official PENGWIN label ranges and pasted into the full label volume:

```
Sacrum fragment k → label k (1–50)
LeftHip fragment k → label k + 50 (51–100)
RightHip fragment k → label k + 100 (101–150)
Femur fragment k → label k + 150 (151–200) ← V0.x pending, V0.3.4 emits 0
```

The volume is reoriented from LPS back to the original input frame before writing the `.mha` output.

7. V0.3.x progression

Tag	Change	Held-out Frac Dice	Inst F1
v0.2	Two-stage anatomy-conditioned pipeline	0.612	0.683
v0.3.1	GC timeout fix (deterministic single-pass)	0.621	0.805
v0.3.2	False-femur suppression (internal)	0.628	0.812
v0.3.3	Mask cleanup + TTA z-flip + median merge	0.634	0.819
v0.3.4	**Bone-skeleton-aware routing + ≤27-vox merge**	**0.641**	**0.826**

Cumulative delta v0.2 → v0.3.4: Fracture Dice +0.029, Instance F1 +0.143, HD95 –8.6 mm.

All v0.3.x tags share the same V291 fold0 checkpoint — only post-processing / routing changes.

8. Hardware + runtime

• Inference GPU: NVIDIA T4 16 GiB (Grand Challenge default).

• Per-case runtime: typically 3–6 min, capped at 480 s (8 min).

• Container image: `pengwin/pytorch:2.0.1-cuda11.8-cudnn8-runtime` + nnUNetv2 2.5.1 + SimpleITK 2.3.1.

9. Limitations

• **Femur not modeled (V0.3.4).** Femur-only cases (50% of the cohort) score zero on those instances. V0.5 adds femur via Dataset532 5-class re-training + 4-anatomy V291 retraining.

• **Sub-fulltrain probe.** Only 12 of 174 training cases used for the V291 fold0 checkpoint shipped in v0.3.4.

• **Single fold, single seed.** No ensembling.

10. References

• nnU-Net v2: Isensee, F. et al. *Nat Methods* 18, 203–211 (2021).

• PENGWIN 2026 Task 1 baseline:

• PENGWIN 2024 1st place (ABBC reference): MIC-DKFZ, IoU-F 0.9296.